श्रद्धावान् लभते ज्ञानम्

Fiduciary Lens

Group No: 11

Category: Software

FACULTY GUIDE

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LinkedIn Product Demo: [URL]

GitHub Repository: https://github.com/Key-to-Code/Fiduciary-Lens-Tax-Advisor-System

This presentation is prepared for ***Course: 23CSE471 – Natural Language Processing***

Problem Statement and Objectives

Area	Detail
Problem	Tax documents use technical terminology, statutory cross-references and figures that ordinary taxpayers cannot interpret without professional help.
Information gap	General AI assistants answer tax questions fluently but give the user no way to check the section number or the limit they quote.
Why it matters	A stale or invented rate reads exactly like a correct one, and rules change every year through the Finance Act.
Objective	Return the governing provision in plain English with a citation the user can verify against the published statute.
Sub-objective 1	Extract text and financial entities from tax documents using PyMuPDF, regular expressions for fixed-shape identifiers, and an LLM for open-ended entities.
Sub-objective 2	Retrieve the relevant provisions with hybrid dense and lexical search over the Income-tax Act 2025, Rules 2026 and Finance Act 2026.
Sub-objective 3	Enforce grounding in code: refuse before calling the model when retrieval similarity falls below threshold, and keep the scope educational.

Similar Products in the Same Area:

Product Name	URL	Features	Limitation
FlyFin	https://flyfin.tax/	AI-powered deduction detection tailored for freelancers; continuous background scanning of bank statements; includes CPA verification of returns.	Geared exclusively towards freelancers and independent contractors, lacking robust tools for complex corporate structures or standard salaried employees (W-2/PAYE).
Keeper (Tax)	https://www.keepertax.com/	Utilises machine learning to automatically discover tax write-offs from linked bank accounts; generates audit-ready documentation and one-click tax reports.	Operates primarily as an expense tracker rather than a comprehensive document explainer; requires linking live bank data, which may raise privacy concerns for some users.
TaxDome	https://taxdome.com/	Comprehensive practice management for accounting firms; features AI workflow automation, automated document collection, and secure client portals.	Designed as a B2B platform for tax professionals rather than a consumer-facing educational tool; highly complex and expensive for individual taxpayers.

Abstract and Novelty

Aspect	Detail
What it is	A grounded tax advisory system that answers questions about Indian income tax only from retrieved statutory text, and analyses a taxpayer's own documents.
Document pipeline	PyMuPDF text extraction, LLM summarisation, regex extraction of PAN / TAN / GSTIN / amounts, and LLM extraction of organisations and deduction sections. No OCR stage: the source PDFs carry a text layer.
Retrieval	BAAI/bge-small-en-v1.5 dense embeddings in FAISS, fused 65/35 with BM25, deduplicated by provision, top 6 passages passed to the generator.
Novelty 1	Grounding is an architectural property, not a prompt instruction: a cosine threshold decides before the LLM is called, so the model cannot override it.
Novelty 2	A refusal guard that retires itself, reading the indexed documents rather than a hardcoded flag, so adding a Finance Act removes the rate restriction automatically.
Novelty 3	Statutory structure recovered after chunking, so every passage carries the section, rule or schedule it came from and provision-level citation becomes possible.
Novelty 4	Document findings closed back into the law: each extracted deduction is re-queried through the same grounded path and arrives attached to its provision.
Scope	Educational only. The system explains what the law says; it does not compute a liability or recommend what to claim, and routes personal decisions to a Chartered Accountant.

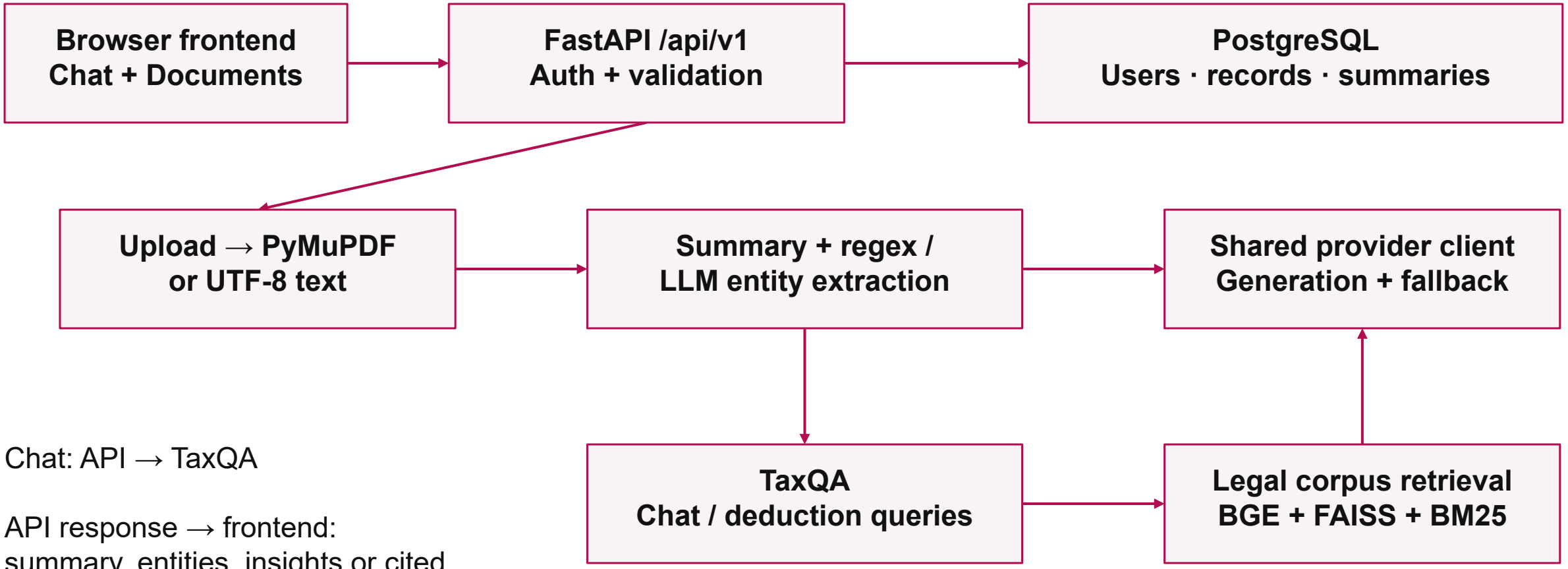
Current Implementation and Features

User capability	Connected implementation
Ask a tax-law question	POST /ask → hybrid retrieval → cited answer + sources
Analyze an uploaded document	POST /documents/analyze → summary + entities + tax insights
Inspect retrieved provisions	Source excerpts, labels and available source links in the UI
Register and sign in	bcrypt password hashing + JWT; document ownership checks
Retain document records	PostgreSQL metadata + summaries; list/detail/delete APIs

Technology Stack and Development Environment

Layer	Verified technology / declared requirement
Frontend	HTML5, CSS3, vanilla JavaScript; static HTTP server
API and schemas	Python; FastAPI ≥0.115; Uvicorn ≥0.30; Pydantic Settings ≥2.3
NLP and retrieval	Sentence Transformers ≥3.4; FAISS CPU ≥1.10; scikit-learn ≥1.5
Document preparation	PyMuPDF ≥1.24; LangChain Text Splitters ≥0.3
Generation	OpenAI SDK ≥1.40; Transformers ≥4.56; Requests ≥2.31
Persistence and access	PostgreSQL; SQLAlchemy ≥2.0; Alembic ≥1.13; bcrypt; PyJWT
Development / deployment	Git, pytest; Uvicorn + static frontend; deployed to a free online server (URL in the readme)

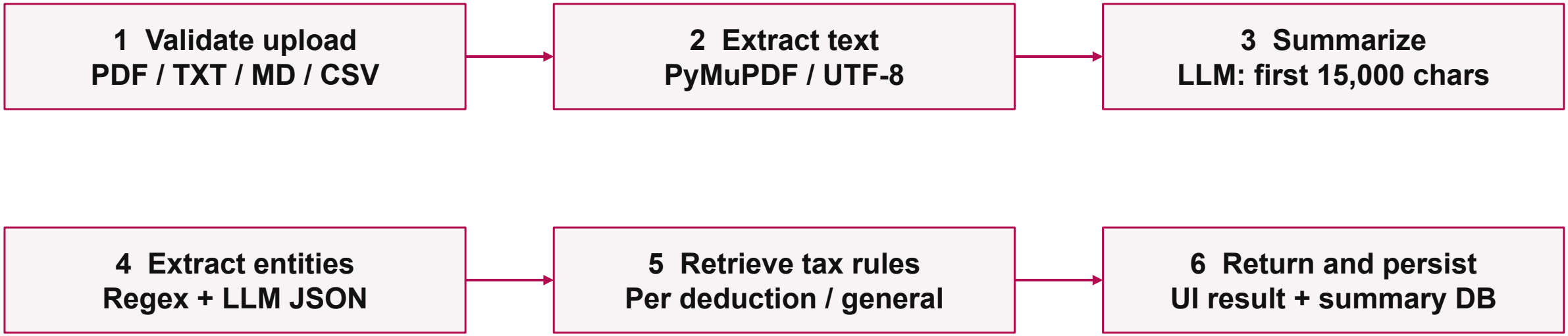
Complete System Architecture



Chat: API → TaxQA

API response → frontend: summary, entities, insights or cited answer

Document Processing and NLP Pipeline



- Regex: PAN, TAN, GSTIN and amounts; LLM: organizations and deduction sections.
- JSON response is cleaned and parsed; related provisions include citation metadata.
- No OCR or trained NER model. No document cleaning or document-vector indexing stage.

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LLMs Used and Model Integration

Exact model / backend	Role and integration status
BAAI/bge-small-en-v1.5 BAAI · Hugging Face	Default retrieval encoder via SentenceTransformer; passages / question → normalized vectors
meta-llama/llama-3.3-70b-instruct Meta · compatible API / OpenRouter	Default configured generator; document summaries, JSON entities and grounded answers; execution unverified
llama3.1:8b · Ollama Qwen/Qwen2.5-1.5B-Instruct · Hugging Face	Alternative implemented adapters; local availability unverified. Extractive mode quotes sources without an LLM.
SLM (from scratch) · BERT-BiLSTM-CRF	Benchmarked for the deep-learning comparison on statutory-source classification; all metrics shown in the application's Evaluation page

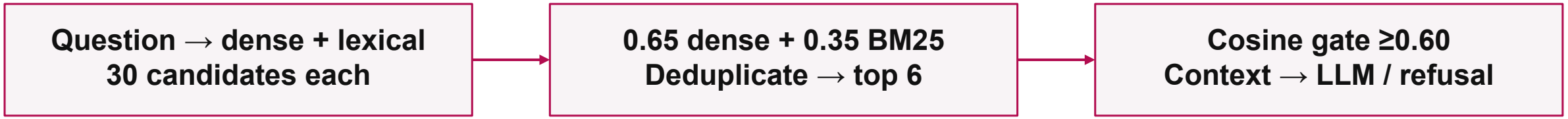
- Prompts: accountant summary; JSON-only entity extraction; numbered legal context for Q&A.
- Q&A defaults: temperature 0.1; max 800 output tokens; last 3 history turns as context.
- No fine-tuning. IndicBERT appears in stale documentation, not as the code default.

Hybrid RAG and Knowledge Retrieval

Offline preparation



Online question answering



- Corpus: Income-tax Act 2025, Income-tax Rules 2026 and Finance Act 2026.
- Numbered sources and excerpts support explanations; generation failure uses quoted provisions.
- “Grounded” indicates sufficient retrieval similarity; citation correctness still needs validation.

End-to-End Working of the Application

- 1 Sign in / register, then select a synthetic Form 16 in Documents.
- 2 Browser sends multipart file + bearer JWT to /documents/analyze.
- 3 API extracts text; provider generates summary and complex entity JSON.
- 4 Detected deduction sections become TaxQA questions over the legal corpus.
- 5 UI shows summary, entities, insights and sources; DB saves record + summary.

Separate chat path: enter a tax question → /ask → hybrid retrieval → cited explanation.

SDG Mapping

Item	Detail
Primary goal	SDG 4 – Quality Education
Targets	4.4 relevant skills for decent work; 4.6 literacy and numeracy for youth and adults. Financial and legal literacy is the competence this product addresses.
How it maps	India's tax statutes are already public and free. What blocks a taxpayer is not access but comprehension. The system is a translation layer that returns the relevant provision in plain English and defines legal terms on first use.
The citation as pedagogy	An answer that merely asserts the right limit teaches nothing. An answer that shows the passage it came from invites the user to check it, and checking is the skill.
Secondary goal	SDG 16, target 16.10 on public access to information. The contribution is traceability rather than access: the user can audit an answer instead of trusting it.
Honest limit	Narrow scope. Three statutes, educational framing, no filing service and no substitute for professional representation.

Retrieval and Grounding Results

Metric	k = 1	k = 3	k = 6 (operating point)
Recall@k	85% (17/20)	100% (20/20)	100% (20/20)
MRR	0.850	0.917	0.917
Refusal rate (off-topic)	100% (5/5)	100% (5/5)	100% (5/5)
On-topic best cosine	0.73 – 0.84 (mean 0.776)		
Off-topic best cosine	0.49 – 0.52 (mean 0.501)		
Threshold 0.60	Sits in an empty band: 0.13 below the weakest on-topic question, 0.08 above the strongest off-topic one.		
Caveat	20 hand-authored questions and 5 controls on the current index. A regression check, not a general performance claim.		

Deep Learning Model Comparison

Metric	Trigram SLM	BERT (probe)	DistilBERT (probe)
Accuracy	0.884	0.849	0.814
Precision (macro)	0.888	0.851	0.819
Recall (macro)	0.884	0.849	0.814
F1 (macro)	0.884	0.850	0.816
Perplexity	552.94	not applicable	not applicable
BLEU / ROUGE-L	0.0361 / 0.1321	not applicable	not applicable
Parameters	91,973 counts	109.48 M	66.36 M
Training time	0.15 s (CPU)	35.18 s (GPU)	8.55 s (GPU)
Inference	0.38 ms/passage	23.51 ms/passage	18.21 ms/passage

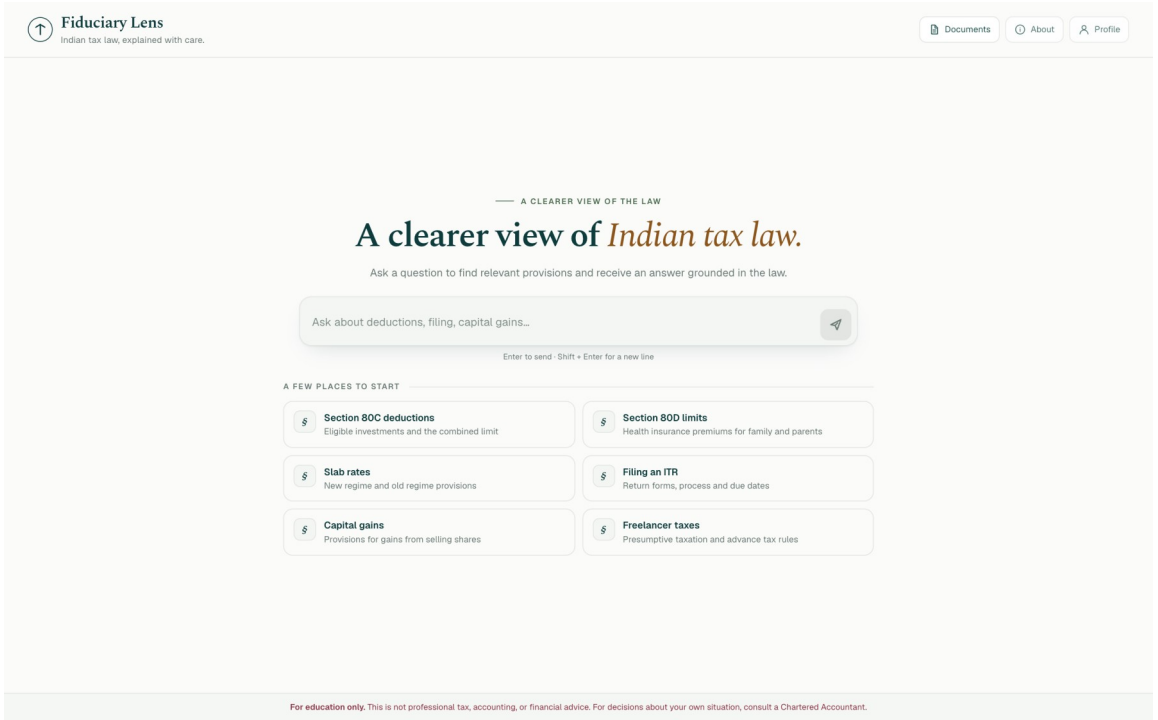
Model Comparison: Setup and Inference

Item	Detail
Task	Statutory-source classification: given a passage, decide whether it came from the Income-tax Act 2025, the Income-tax Rules 2026 or the Finance Act 2026. This is the labelling step the citation layer depends on, not a borrowed benchmark.
Dataset	Balanced to 344 passages per class from a raw 1,918 / 1,234 / 344 split, giving 1,032 passages; seeded 75/25 split into 774 train and 258 test.
Preprocessing	NLTK word tokenisation, punctuation and 198 English stop-words removed, WordNet lemmatisation. Lemmatised rather than stemmed so the XAI tokens stay readable.
Models	Trigram SLM written from scratch (class-conditional, add-k 0.1, 8,000-type vocabulary). BERT and DistilBERT used as frozen encoders with a logistic-regression head, not fine-tuned.
Inference drawn	The SLM wins every classification metric using about 1/1190 of BERT's parameters and running roughly 60x faster. The classes are separable on surface vocabulary, which is what an n-gram captures and what a mean-pooled embedding discards.
Why it does not generalise	The transformers were probed, not fine-tuned, and an n-gram model cannot do the paraphrase matching the retrieval path depends on. Match the model class to where the signal lives; this is the same finding that motivated hybrid retrieval.

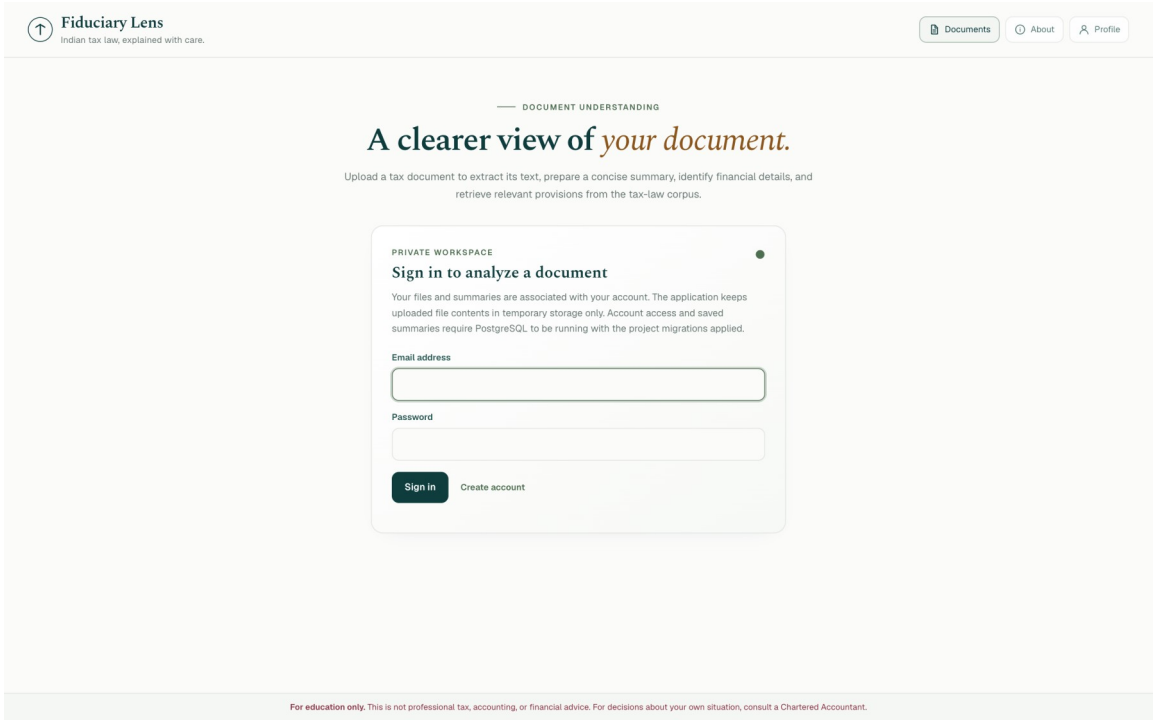
Explainable AI

Mechanism	Detail
1. Hybrid Retrieval Score Attribution	Every retrieved passage reaches the user with its score decomposed rather than as one opaque number.
Formula	score = (dense weight × normalised cosine + BM25 weight × normalised BM25) × statute weight
Why it helps	A passage retrieved mostly on lexical match means the system found your exact section number. A high dense and near-zero lexical score means it matched meaning, not wording, which is when a human should read the passage more carefully.
2. Occlusion attribution	Leave-one-token-out. Each token is removed, the model re-scored, and the drop in the margin between the predicted class and its closest rival recorded as that token's contribution.
Why occlusion	Model-agnostic, so one method covers all three classifiers including the count-based SLM, which has no gradients. This is what LIME approximates with a surrogate, computed exactly.
Worked example	Passage classed as Rules (true: Rules). Top tokens: harbour (+0.146), assessee (-0.130), loan (+0.114), research (+0.095), 88. (+0.086)
Finding	The decisive tokens are structural and procedural vocabulary, not semantic content, which is exactly why the shallow model competes with the transformers on this task.

Product Demonstration and Outputs



Tax-law chat: question entry and topic shortcuts



Documents: sign-in gate before file analysis

Output contracts: answer + sources + grounded; summary + entities + tax_insights.

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Deployment, Dataset and Submission Links

Item	Link / status
Hosted application	<DEPLOYED URL – fill in before submission>
GitHub repository	https://github.com/Key-to-Code/Fiduciary-Lens-Tax-Advisor-System
IEEE DataPort dataset	https://ieee-dataport.org//documents/fiduciary-lens-dataset
Reference journal paper	https://www.ijsr.net/getabstract.php?paperid=ES24115141827
Demo recording	<GOOGLE DRIVE LINK – fill in before submission>
Corpus	Income-tax Act 2025, Income-tax Rules 2026, Finance Act 2026. 3,496 passages, 1,200 characters with 250 overlap.
Reproducibility	python rag_model/scripts/build_index.py → index; python rag_model/scripts/eval_retrieval.py → retrieval metrics; python nlp_pipeline/src/model_benchmark.py → the three-model comparison.

Member Contributions

Roll No	Name	Technical contribution
CB.SC.U4CSE24107	Anuja R S	Backend APIs and system integration connecting document processing, the AI modules and the frontend
CB.SC.U4CSE24118	Ganesh Sarathi	User interface for document upload, financial summaries and taxpayer-friendly insights, including the evaluation and members pages
CB.SC.U4CSE24119	Gaurav Chandrasekhar	AI and NLP pipeline for financial entity extraction, document summarisation and tax insight generation
CB.SC.U4CSE24151	Sudharshan Sambathkumar	Tax-law dataset construction and preprocessing from official sources, retrieval index and evaluation
Guide	Dr. Senthil Kumar T.	Professor, Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore